CSE 12 - Winter 2026 Syllabus

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- Course schedule (lecture contents, discussion materials, and PAs): link here
- Course discussion forum: www.piazza.com
- PA submission: www.gradescope.com
- Tutor help session: autograder.ucsd.edu
- Gradebook and course materials: canvas.ucsd.edu

Some general links for your reference (not specific to CSE 12)

- Undergraduate Course Updates
- Advising / mentorship for all students in CSE classes from CSE's UGCom
- CSE Undergraduate Student Opportunities and Support

General Information

1. Prerequisites

This course expects that you have successfully completed CSE 8B or CSE 11 at UCSD. We assume comfort and skill with basic Java programming, including using and creating objects, drawing memory models, developing and programming algorithms to manipulate strings, arrays, and numerical data, writing programs that read and write from and to files, and building small to moderate size Java programs using the Java API as a reference.

If you have completed a rigorous introductory class in another object-oriented language (e.g., C++), you are probably OK to take this course, but you must learn the basics of Java on your own BEFORE this course begins. If you have taken an introductory course in a different language (Python, C), we recommend you take CSE 11 or CSE 8B before this course.

2. Critical information, at a glance

You should read this entire syllabus. It is important. It may be the most important thing you read for this course. But here are the pieces of information you absolutely do not want to forget. I don't mean for this to sound scary, but so many students fail to read or understand these points, so I want to make them as clear as possible.

- Homework (PAs) is due by 11:59 pm on the due date. No late work will be accepted unless it is due to a documented emergency, and the instructor of the course has to approve it before the due time. Unless you select otherwise on Gradescope, we only grade your most recent Gradescope submission.
- All reading assignments are due before 11:59 pm on the Friday of the week. For example, week 2 readings are due before 11:59 pm on Friday of week 2. Late readings will lose 10% per day after the deadline.
- It is your responsibility to ensure that you have correctly submitted the correct code for your homework assignment. Incorrectly submitted assignments will be graded as is.
- All homework assignments must be done based on the instructions.
- All questions for the class should be posted to Piazza. Emails to the instructor should be about personal and confidential matters only.
- We allow students to resubmit their work up to 1 week after you receive your PA grade and receive up to 50% of their lost points in the auto-graded portion. More details on the redo part of the PAs in the course components part of the syllabus.
- Our midterm and final exam will include two parts: MCQ and coding. You must score at least **55% on the coding portion** of the final exam to pass the course.

3. What will I learn in this class?

In this course, you will continue to develop your sophistication as a programmer by learning to write larger-scale programs that use and manage data efficiently and correctly. You will learn several basic ways of storing data (data structures), how to analyze your programs and your data structures, and how to use abstraction appropriately to create efficient, correct, and reusable programs.

This course is more challenging than CSE 8B and CSE 11 in that your projects will be slightly larger, you will be expected to do more of the design (and analysis) on your own, and you will be given more freedom in your implementations. However, we expect that everyone who has succeeded in CSE 8B or CSE 11 can succeed in this course too.

Here are some detailed topics

- Java Collections Library
- The use of generics, interfaces. abstraction, ADTs, invariants
- Exception design and handling
- Runtime analysis
- List structures: ArrayList and LinkedList
- Hashing technique
- Stack and Queue
- Heap and Priority Queue
- Binary Tree and Binary Search Tree
- Sorting algorithms (comparison based and non-comparison based)

4. Textbook: zybooks.com for the e-book.

- Click any ZyBooks assignment link in your learning management system. **Do not go to the zyBooks website and create a new account.**
- Subscribe

We assign reading from the book as well as some in-session exercises. **You must access all ZyBooks assignments via Canvas initially so that your score will be recorded in Canvas.**

Staff and Course Information

1. Course Staff

Instructor: Paul Cao

Contact: EBU 3B 2102, yic242@ucsd.edu

Office hours: Mondays 10am - 11am, Thursdays 2pm - 3pm in CSE 2102.

If you cannot make it to the scheduled office hours, email the instructors to set up an appointment. Please only email instructors with personal (confidential) questions. If you need to see us at a time other than office hours, you should email us in advance to set up a time. You should look at their calendar and propose a few time slots of 30 minutes or less.

TAs

Our TAs will host their office hours in the CSE building. Please refer to their office hours on the autograder (see tutors). You don't have to submit tickets to get TA help during office hours. It is simply first come, first serve.

- Kristian Ortiz, klortiz@ucsd.edu Office Hours:
  - Mondays, 4:00 PM - 6:00 PM, in CSE B215.
  - If remote, use this zoom link https://ucsd.zoom.us/j/8148992572
- Luyi Li, lul014@ucsd.edu, Office hours:
  - Tuesday, 3:30 PM - 5:30 PM, in CSE B215
- Trey Shneour, tshneour@ucsd.edu, Office hours:
  - Wednesdays, 2:30 PM - 4:30 PM, in CSE B215

Tutors

We have many tutors for this class who are available to help you online. If you need tutor help, just submit your ticket on autograder.ucsd.edu. Our default home base for tutoring is the CSE building basement. Please try to be in the CSE basement when requesting tutor help to minimize wait times.

2. Course Components

1. Class Sessions (a.k.a. Lectures)

These are instructor-led sessions that will focus on explanations of complicated and important concepts, and explore strategies on manipulating basic data structures. Instructors will use in-class polls as well as pre-released worksheets to work out different problems with students. These sessions will be podcasted. It is strongly encouraged that everyone attends lecture sessions for more effective learning.

We will be using a tool called **Webclicker** as a polling system during lecture (https://webclicker.web.app/).

Once on the Webclicker page, click on the button "SIGN IN WITH GOOGLE". Make sure to use your UCSD email address (i.e., @ucsd.edu) to sign in. If you use another Google email address, you will not receive course participation credit.

If everything goes well, you will now be logged into the tool. Enter **UFRZMG** in the course code field. You will now be able to select **cse12-cao**.

During lecture, you can vote on the polls by logging into the Webclicker site with your phone, laptop, or computer, and selecting our course. Once we start a poll, you will see the voting buttons appear.

2. Reading Assignments

To prepare you for these interactive class sections, there will be reading assignments for each class section. There are also videos that accompany the reading. Though the due time is Friday 11:59 pm of each week, it is strongly recommended that you complete the reading assignments before each lecture.

3. Exams and Quizzes

We have one midterm, three quizzes (given in the discussion section), and a comprehensive final exam. We will provide more details about the exams later in the quarter.

4. Programming Assignments (PAs)

Most weeks, there will be a problem-solving assignment due by 11:59 pm on the due date. These will consist of reading a problem statement, formulating a plan to solve the problem (an algorithm), and implementing a solution (writing a program in Java that solves the problem). **Unless you manually select an older submission, we will only grade your most recent submission.** Make sure that your submission runs without any compiler or autograder errors. You can always rollback to an older submission before the deadline by clicking the “Submission History” button on Gradescope and selecting a previous submission as your active submission.

After a PA deadline has passed, we will enable a re-do submission window (usually a week) where students can submit their modified assignment again to improve their grade. Students can get up to 50% of their lost points back. **For redo submissions only, we encourage our students to use AI tools** to develop some prompt engineering skills and experience potential efficiency improvements with the help of AI tools. We will not give you any immediate feedback on resubmission assignments. This includes compile checks, import checks, and test cases that were previously public on the initial assignment. **No AI tools are allowed for initial submissions of any assignment.**

You do not need to spend money on any AI tools, and doing so will not give you any significant advantage. https://tritongpt.ucsd.edu/ is provided to all UCSD students at no cost. If you use TritonGPT, ensure that you click "General AI Assistant" on the left sidebar. If you give it explicit permission, then it should work (if it does not, please make a Piazza post).

Other free student resources (not an endorsement):

- Google Gemini
- Cursor

5. Discussion Sections

We use discussion sessions to conduct our quizzes, and it is important that you make plans to attend the discussions. We will also address the answers to the quizzes after the quizzes are due during the discussion session. Details can be found in the course schedule.

Grading Policy

Your final grade will be determined via the following percentages:

- Reading assignments from ZyBooks (drop 1 week): 10%
- Class participation (drop 2 weeks of participation and start to count participation starting week 2): 5%
- Quizzes: 10%
- Midterm: 10%
- PA: 40%
- Final exam (a portion of the final can be used to substitute the midterm): 25%

We will use a standard scale for assigning letter grades:

- A>=90
- B: 80-89.99999
- C: 70-79.99999
- D: 60-69.99999
- F:<60

The instructor has the discretion to assign + or - for a given letter grade, though in general, we will stick with the normal scales for + or -. 25

Important Grading Policies:

- **You must score at least 55% on the overall PA portion of this course. You must also score at least 55% on the coding portion of the final exam.** If you score lower than the 55% cutoff on the aforementioned categories, you will receive an F for the course, regardless of your overall average.
- You have **3 days from the time a PA or in-class quiz is returned to request a regrade.** After that, the grade is set in stone. You should submit your regrade request through Gradescope. Submit the regrade request on the corresponding problem and clearly explain why you think there is a grading error. If you submitted a regrade request without clear justification and were simply abusing the regrade system, **your grade will be lowered as a penalty.**
- The **last PA** and the **final exam** have a **regrade period of 1 day after the work is released to you.** It will ensure that we submit everyone's final letter grade on time. No regrade requests can be made after the regrade period has ended.

Policy on late work, missed quizzes, making up work, etc:

- PAs should be submitted by the deadline, and we won't accept late work unless it is due to a documented emergency. You should contact a TA or the instructor if an emergency has happened.
- Each quiz will have a corresponding makeup quiz a week later. Anyone can take the makeup quiz, and **the higher of the original and the makeup quiz will be the final score for that quiz.**
- Late readings will lose 10% per day after the deadline.
- We allow students to resubmit their work up to 1 week after you receive your PA grade and receive up to 50% of their lost points in the auto-graded portion.

If you need accommodations beyond what is listed above, please contact your instructor ASAP. We are happy to make arrangements for further extensions or adjustments for students facing extreme circumstances.

Academic Integrity and Collaboration

Complete your academic honor pledge in this link before the end of Week 1!

1. Academic Integrity

The goal of CSE 12 will be a turning point for most of you in understanding how different strategies to solve a problem may result in vastly different efficiency. Future instructors, employers, and colleagues will expect that you will attain a certain proficiency in data structures. Using unauthorized aids in doing your work will prevent you from attaining the proficiencies that others will expect. Meeting or not meeting these expectations relate directly to getting internships, retaining employment, and success in your future studies.

The basic rule for CSE 12 is: Work hard. Start early. Make use of the expertise of the CSE 12 staff to learn what you need to know to really do well in the course. Don't cheat.

If you do cheat, we will enforce the UCSD Policy on Integrity of Scholarship (see the General Catalog or the corresponding web page, scroll 2/3 of the way down). This means: You will get a reduced grade in this course (in many cases, an F), and the Dean of your college could put you on probation or suspend you, or dismiss you from UCSD.

What counts as cheating?

In CSE 12, you can read books, surf the web, talk to your friends and the CSE 12 staff to get help understanding the concepts you need to know to solve your PA problems. However, you must write your program alone for all PAs except where otherwise indicated. Specifically:

On PAs, you MAY:

- Consult any resources provided in class or in the textbook.
- Ask tutors, TAs, and instructors for help.
- Search/consult the web for general documentation, for example, how to use Java APIs.
- Use AI tools on the **resubmission** portion of the PA

On PAs, you may **NOT**:

- Search/consult the web for solutions to the PAs, including algorithms or code.
- Use AI tools for the **regular** submission of the PA
- Have anyone else completed your assignment (or any part of it).
- Post code in a public GitHub repository or fork the starter code.
- Copy JUnit test files from someone else

Note: This is not a fully inclusive list.

We do electronically check every program that is turned in. In recent quarters, we also found out that people unintentionally post their codes on public GitHub repos and it is a violation of the AI policy! We report all these cases to the academic integrity office. If you want to share your work with future employers or simply want to save a copy of your work, you can post them privately on your GitHub account! **Any public repo with your work in it will be treated as a violation of academic integrity.**

Receiving a grade on a PA doesn't mean that you have passed the plagiarism checking. We can report cheating cases any time during the quarter, even after we submit your final letter grade. So the safest bet is don't cheat!

To reinforce these points, every student in CSE 12 must read, understand, and sign the Integrity of Scholarship Agreement. You must sign an integrity statement before completing any of the assignments or labs.

2. Getting Help

We expect that ALL STUDENTS WILL NEED HELP at some point in this course. If you find yourself needing help, this is not a cause for embarrassment. It is completely expected. Please seek help early and often through any (or all) of the following resources.

Open lab hours:

There are many, many hours in which tutors are willing and available to help you with any questions you might have. The course tutor calendar can be found on the first page of the syllabus.

Online help - Piazza

Use the piazza site for any questions related to the PA, material in the course, quizzes, exams, or course logistics. Piazza allows you to post questions anonymously (to other students) if you don’t feel comfortable revealing your name. In general, all questions should be addressed to Piazza EXCEPT:

- Confidential questions that you wish to address to your instructor only. Please email your instructor directly.
- Posts that include portions of your PA solution. Please see a tutor during our tutoring session or post these questions privately.

Office hours:

Your instructor and TAs are always willing to help you during their office hours. If you cannot make it to your instructor's or TA's office hours but wish to meet with your instructor or TA, please send them an email to set up an appointment.

Other Important Policies

1. Diversity and Inclusion

We are committed to fostering a learning environment for this course that supports a diversity of thoughts, perspectives, and experiences, and respects your identities (including race, ethnicity, heritage, gender, sex, class, sexuality, religion, ability, age, educational background, etc.). Our goal is to create a diverse and inclusive learning environment where all students feel comfortable and can thrive.

Our instructional staff will make a concerted effort to be welcoming and inclusive to the wide diversity of students in this course. If there is a way we can make you feel more included please let one of the course staff know, either in person, via email/discussion board, or even in a note under the door. Our learning about diverse perspectives and identities is an ongoing process, and we welcome your perspectives and input.

We also expect that you, as a student in this course, will honor and respect your classmates, abiding by the UCSD Principles of Community (https://ucsd.edu/about/principles.html). Please understand that others' backgrounds, perspectives, and experiences may be different than your own, and help us to build an environment where everyone is respected and feels comfortable.

If you experience any sort of harassment or discrimination, please contact the instructor as soon as possible. If you prefer to speak with someone outside of the course, please contact the Office of Prevention of Harassment and Discrimination: https://ophd.ucsd.edu/.

2. Students with Disabilities

We aim to create an environment in which all students can succeed in this course. If you have a disability, please contact the Office for Students with Disability (OSD), which is located in Pepper Canyon Hall Suite 300, to discuss appropriate accommodations right away. We will work to provide you with the accommodations you need, but you must first provide a current Authorization for Accommodation (AFA) letter issued by the OSD. You are required to present their AFA letters to the Faculty (please make arrangements to contact me privately) and to the OSD Liaison in the department in advance so that accommodations may be arranged.

If you need extra time for quiz and exams, please contact the Triton Testing Center to get the accommodations you need. Your AFA letter should have included the information for Triton Test Center information. Make sure you make appointments with TTC as early as you can so you can secure a spot. Please do not go to class time to do the quiz or exam as we cannot make accommodations in our regular classrooms, as there may be a class right after our class.

3. Basic Needs/Food Insecurities

If you are experiencing any basic needs insecurities (food, housing, financial resources), there are resources available on campus to help, including The Hub and the Triton Food Pantry. Please visit http://thehub.ucsd.edu/ for more information.

4. Late Add Policy

The CSE department (https://cse.ucsd.edu/undergraduate/courses/enrolling-cse-courses) does not allow students to add courses past week 2. If a student is not enrolled in enough units by the beginning of week 1, the student should contact their department advisors via the VAC.
No late adds - no exceptions. EASy requests for Late Adds in the CSE Department will be denied, so please plan accordingly.

Please note, all students are expected to attend class for the first two weeks and complete assignments if they are on the waitlist for a course. Attending class and completing course assignments does not guarantee enrollment. If students choose to miss class or not turn in assignments while on the waitlist, the student will receive a "0" on all missed assignments, if they secure a seat in the course off the waitlist.

5. Incomplete

Sometimes, circumstances beyond a student's control prevent them from completing a class even once they have completed the majority of the coursework at a passing level. UCSD has a process in place for you to request an Incomplete (I) if this happens to you. Here is the campus policy about the Incomplete grade:
https://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Regulations/500 and some information about it:
https://students.ucsd.edu/academics/exams-grades-transcripts/grades/request-remove-incomplete.html

6. Student Conduct Policy

UC San Diego strives to maintain a climate of fairness, cooperation, and professionalism. It is expected that you practice basic principles, including, but not limited to, mutual respect, civility, and decency, towards maintaining an atmosphere free of abusive or demeaning treatment. Non-academic student misconduct will be reported to the Center for Student Accountability, Growth, and Education for violating UC San Diego's Principles of Community.

7. Outside Tutoring

Individuals are not permitted to approach students to offer services of any kind in exchange for pay, including tutoring services. This is considered solicitation for business and is strictly prohibited by University policy.

8. Class material and intellectual property

Our lectures and course materials, including videos, assignments, and similar materials, are protected by U.S. copyright law and by University policy. We are the exclusive owner of the copyright in those materials we create. We acknowledge the cumulative contributions to this course material of previous course instructors, TAs, and tutors, as well as contributions to the class structure from colleagues in CSE and at UCSD.

You may take notes and make copies of course materials for your own use. You may also share those materials with another student who is enrolled in or auditing this course. You may not reproduce, distribute or display (post/upload) lecture notes or recordings or course materials in any other way — whether or not a fee is charged — without our express prior written consent. You also may not allow others to do so. If you do so, you may be subject to student conduct proceedings under the UC San Diego Student Code of Conduct.

Similarly, you own the copyright in your original work. If I am interested in posting your answers or papers on the course website, I will ask for your written permission.

9. Satisfactory Academic Progress

Satisfactory Academic Progress (SAP) refers to the academic standards students must maintain to remain eligible for federal, state, and institutional financial aid. If you are receiving financial aid, please ensure you review the SAP requirements and the appeals process. More details at this campus website: https://fas.ucsd.edu/forms-and-resources/sap/index.html.